Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01]

Choose the solid model before computing: similarity controls scale, a perpendicular section controls height, and coordinates control clipping. Surface area, volume and surface path length require different decompositions. Every additive calculation must respect overlap and newly created faces.

先确定立体模型，再计算：相似控制缩放，垂直截面控制高度，坐标控制截割。表面积、体积与表面路径需使用不同分解；相加之前必须核实重叠及新产生的面。

Similarity has different area and volume exponents

相似的面积与体积具有不同指数

Procedure

1. Identify a common linear scale t between corresponding solids.

2. Use area scale t^2 and volume scale $|t|^3$; for a complementary piece subtract from the whole.

3. Distinguish inherited paint from a newly cut face, and check $0 < t < 1$ for a true subdivision.

通用步骤

1. 确定对应立体的线性相似比 t。

2. 面积比为 t^2，体积比为 $|t|^3$；补块须从整体相减。

3. 区分原有油漆与新切面，真正分割须满足 $0 < t < 1$。

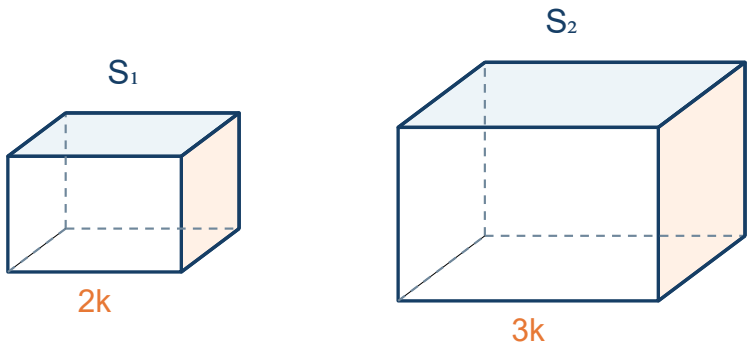
> Example. Two similar closed boxes have corresponding edge lengths in ratio $2:3$. Their total surface area is 130. Find each surface area and the ratio of their volumes.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01] (continued / 续)

> 例题：两个相似的封闭长方体，对应棱长比为 $2:3$，表面积之和为 130。求各自表面积及体积比。



S₁ + S₂ = 130 edge ratio / 棱长比 = 2 : 3

Worked solution

1. Area scales as the square of the linear ratio, so write the two surface areas as $4c$ and $9c$.

2. The total gives $13c = 130$, hence the areas are 40 and 90.

3. Volume scales as the cube, so the volume ratio is $8:27$. Similar boxes realizing these areas exist by uniformly scaling any fixed box.

例题解析

1. 面积按线性比的平方缩放，故把两表面积写成 $4c, 9c$。

2. 总和给出 $13c = 130$，所以表面积分别为 $40, 90$。

3. 体积按线性比的立方缩放，故体积比为 $8:27$。将任一固定长方体整体缩放，即可实现这些表面积。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01] (continued / 续)

A fresh cut surface carries no original paint.

新切面没有原来的油漆。

For a tetrahedron with centroid G, the opposite face centroid is $G-(A-G)/3$; the four face centroids form a tetrahedron of volume $1/27$ of the original.

四面体重心为 G 时，顶点 A 对面重心为 $G-(A-G)/3$；四个面重心形成的四面体体积为原来的 $1/27$。

Practice / 配套练习: AIME-2000-I-Q08 (WB p. 444), AIME-2003-II-Q04 (WB p. 438), AIME-2004-I-Q11 (WB p. 453)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 2 / 方法 2 [GE10-M02]

Recursive face counts require non-overlap certification

递归面数需要不重叠证明

Procedure

1. Track the number of exposed faces and their common linear scale at each stage.

2. Multiply the number of newly attached copies by the cube of their scale, forming an exact increment sequence.

3. Before summing, prove distinct attachments have disjoint interiors and the counted faces remain exposed.

通用步骤

1. 逐阶段记录外露面数及其共同线性尺度。

2. 新附加副本数乘以尺度立方，得到精确体积增量数列。

3. 相加之前，先证明各附加块内部不交，所计表面仍然外露。

> Example. Attach one outward square pyramid of height $1/2$ to each face of a unit cube, using that entire face as its base. Find the new volume and justify adding the six pyramid volumes.

> 例题：在单位立方体每个面外侧，各接一个以整面为底、高为 $1/2$ 的正四棱锥。求新体积，并说明为何可将六个棱锥的体积相加。

Worked solution

1. Each pyramid has base area one and volume $1 \cdot (1/2)/3 = 1/6$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 2 / 方法 2 [GE10-M02] (continued / 续)

2. The pyramid on $x = 1$ lies in $x \geq 1$, with its other coordinates in $[0, 1]$; the corresponding statement holds at every face. Distinct outward pyramids therefore have disjoint interiors and meet the cube only on its boundary.

3. The total is $1 + 6(1/6) = 2$. The geometric separation, rather than the number of faces alone, justifies the sum.

例题解析

1. 每个棱锥底面积为一，体积为 $1 \cdot (1/2)/3 = 1/6$。

2. $x = 1$ 面上的棱锥满足 $x \geq 1$，另两坐标均在 $[0, 1]$ 内；其余面同理。不同外接棱锥的内部互不相交，且只在边界接触原立方体。

3. 总体积为 $1 + 6(1/6) = 2$。能相加的依据是几何隔离，而非仅仅知道面数。

A plausible face recursion can fail after self-intersections; do not extend it without proof.

发生自交后，貌似合理的面数递推也可能失效，不能无证明推广。

For convex polyhedra, separating-axis tests use face normals and cross products of edge directions. A disjoint projection interval proves interior separation; exact rational coordinates prevent numerical ambiguity at boundary contacts. For repeated attachments, also test final boundary faces for hidden positive-area overlap.

凸多面体的分离轴可取各面法向及两棱方向叉积。投影区间内部不交即可证明内部隔离；使用精确有理坐标可避免边界接触的浮点歧义。递归附加还需检验最终表面是否存在隐藏的正面积重叠。

Practice / 配套练习: AIME-2001-II-Q12 (WB p. 450)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03]

Coordinate planes recover solid geometry and exposed surfaces

坐标平面重建立体几何与外露表面

Procedure

1. Place known faces in coordinate planes and translate planarity into linear equations.

2. Use equal-distance subtraction or a linear combination of squared distances to cancel unknown coordinates.

3. For a drilled solid, split surface area into retained exterior faces and new tunnel-wall polygons; count both separately.

通用步骤

1. 将已知面放在坐标面内，把共面条件写成线性方程。

2. 通过等距相减或距离平方线性组合消去未知坐标。

3. 对钻孔立体，把表面积分为保留外表面及新隧道壁多边形，分别计数。

> Example. In a cube with vertex $A=(0,0,0)$, let B, C, D be its three neighbors and G the opposite vertex. If $PB^2+PC^2+PD^2-PG^2=50$, find PA.

> 例题：立方体顶点 $A=(0,0,0)$，B, C, D 为三个邻点，G 为对顶点。若 $PB^2+PC^2+PD^2-PG^2=50$，求 PA。

Worked solution

1. Write neighbors as $(s,0,0), (0,s,0), (0,0,s)$, opposite vertex (s,s,s), and $P=(x,y,z)$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03] (continued / 续)

2. Expanding the four squared distances cancels all linear terms and all s^2 terms, leaving $2(x^2+y^2+z^2)=50$.

3. Therefore $PA^2=25$ and $PA=5$, independently of cube side length.

例题解析

1. 把三邻点写为 $(s,0,0),(0,s,0),(0,0,s)$，对顶点为 (s,s,s)，并设 $P=(x,y,z)$。

2. 展开四个距离平方，线性项与全部 s^2 项消去，得 $2(x^2+y^2+z^2)=50$。

3. 因此 $PA=5$，与立方体棱长无关。

Distance to a filled face equals distance to its plane only when the perpendicular foot lies in the face.

只有垂足在面内时，到填满面的距离才等于到所在平面的距离。

A tunnel bounded by planes parallel to a cube diagonal can meet each cube face in different polygons. Solve each plane–face intersection before subtracting exterior openings or adding wall areas.

If a tetrahedron has two faces with exceptional edge length $2a$ and two with exceptional length $2b$, the exceptional edges must be opposite: adjacent exceptional edges would belong to one face together. With four equal cross edges L, use vertices $(\pm a,0,h),(0,\pm b,0)$, where $h^2=L^2-a^2-b^2$. This differs from a tetrahedron whose three opposite-edge pairs are equal.

由平行于立方体体对角线的平面围成的隧道，可能在不同立方体面上截出不同多边形。必须先求各平面与面的交线，才能扣除外部开口并加入隧道壁面积。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03] (continued / 续)

若四面体两面含特殊棱长 $2a$，另两面含特殊棱长 $2b$，两特殊棱必为对棱；若相邻，它们会同时出现在某一面。其余四棱均为 L 时，可取顶点 $(\pm a, 0, h), (0, \pm b, 0)$，其中 $h^2 = L^2 - a^2 - b^2$。此模型不同于三组对棱各自相等的四面体。

Practice / 配套练习: AIME-2001-II-Q15 (WB p. 451), AIME-2002-I-Q15 (WB p. 452), AIME-2021-I-Q06 (WB p. 449), AIME-2026-II-Q12 (WB p. 435)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 4 / 方法 4 [GE10-M04]

Cube-vertex distances have only three types

立方体顶点距离只有三类

Procedure

1. Normalize the cube side to s; vertex-pair squared distances are $s^2, 2s^2, 3s^2$.

2. Use adjacency and opposite-vertex constraints to reject impossible triples of distance types.

3. Count triangles by their squared-side patterns and check their counts against $\binom{8}{3}$.

通用步骤

1. 设立方体棱长为 s，顶点间距离平方仅为 $s^2, 2s^2, 3s^2$。

2. 利用邻接与唯一对顶点关系排除不可能的三边类别。

3. 按边长平方型统计三角形，并与 $\binom{8}{3}$ 核对总数。

> Example. How many equilateral triangles have all three vertices among the vertices of a cube? What fraction of the cube-vertex triples do they form?

> 例题：立方体顶点中有多少组三点能构成等边三角形？它们占全部顶点三元组的比例是多少？

Worked solution

1. Normalize vertices to $\{0,1\}^3$. A pairwise equal-distance triple cannot use cube edges or space diagonals, so all its squared side lengths must be two.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 4 / 方法 4 [GE10-M04] (continued / 续)

2. Partition the eight vertices by the parity of their coordinate sum. Each class contains four vertices; any two distinct members differ in exactly two coordinates.

3. Each parity class therefore forms a regular tetrahedron, contributing $\binom{4}{3} = 4$ equilateral triangles. There are eight altogether.

4. There are $\binom{8}{3} = 56$ vertex triples, so their fraction is $8/56 = 1/7$.

例题解析

1. 将顶点归一为 $\{0,1\}^3$。两两等距三点不能以棱或体对角线相连，所以各边平方只能为二。

2. 按坐标和的奇偶把八顶点分为两类，每类四点；同类不同点恰有两个坐标不同。

3. 所以每类形成正四面体，贡献 $\binom{4}{3} = 4$ 个等边三角形，共八个。

4. 全部顶点三元组有 $\binom{8}{3} = 56$ 个，所占比例为 $8/56 = 1/7$。

An equilateral vertex triple is not a face triangle.

顶点组成的等边三角形不在某一个正方形面内。

Practice / 配套练习: AIME-2002-II-Q02 (WB p. 436), AIME-2003-I-Q06 (WB p. 445)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 5 / 方法 5 [GE10-M05]

Rounded boxes decompose by the nearest feature

圆角长方体按最近特征分解

Procedure

1. Separate the original box, face slabs, edge quarter-cylinders and corner sphere octants.

2. Group congruent fractional pieces before evaluating their volumes.

3. Use the nearest-feature partition to justify disjoint interiors and complete coverage.

通用步骤

1. 分成原长方体、面上薄层、棱旁四分之一圆柱及顶点八分之一球。

2. 先合并全等的分数块，再计算体积。

3. 按最近特征划分可证明各块内部不交且完整覆盖。

> Example. A solid box has side lengths $a, b, c > 0$. A coating consists of all exterior points at distance at most $r > 0$ from the box. Derive the coating volume as a polynomial in r.

> 例题：实心长方体三边为 $a, b, c > 0$。涂层由长方体外部距其不超过 $r > 0$ 的点组成。把涂层体积写成关于 r 的多项式。

Worked solution

1. The six face slabs have total volume $2(ab + ac + bc)r$; the original box is excluded from the coating.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 5 / 方法 5 [GE10-M05] (continued / 续)

2. The four edges parallel to a side of length a contribute four quarter-cylinders, totaling $\pi a r^2$. The three directions together give $\pi(a+b+c)r^2$.

3. The eight corner octants combine into one sphere of volume $4\pi r^3/3$.

4. The coating volume is $2(ab+ac+bc)r+\pi(a+b+c)r^2+4\pi r^3/3$. Assigning each exterior point to its nearest box face, edge or vertex proves coverage and disjoint interiors.

例题解析

1. 六个面上的薄层总体积为 $2(ab+ac+bc)r$；原长方体不属于涂层。

2. 平行于长度 a 的四条棱各贡献四分之一圆柱，共为 $\pi a r^2$；三个方向合计 $\pi(a+b+c)r^2$。

3. 八个角部八分之一球合成一个球，体积为 $4\pi r^3/3$。

4. 涂层体积为 $2(ab+ac+bc)r+\pi(a+b+c)r^2+4\pi r^3/3$。按外部点的最近面、棱或顶点分类，可证明完整覆盖及内部互不相交。

Adding twice the radius to every box dimension replaces curved edge and corner pieces by rectangular ones and gives the wrong volume.

直接把各边长加上两倍半径，会错误地把曲面角部换成方角。

Practice / 配套练习: AIME-2003-I-Q05 (WB p. 437)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 6 / 方法 6 [GE10-M06]

An affine height averages to its value at the base centroid

仿射高度的平均值等于底面重心处高度

Procedure

1. Write the upper-minus-lower plane separation as an affine height $h(x, y)$.

2. On a centrally symmetric base, pair opposite points: their heights sum to twice the central height.

3. Multiply base area by this average height, checking the height is nonnegative throughout.

通用步骤

1. 将上下平面之差写为仿射高度 $h(x, y)$。

2. 在中心对称底面上配对相反点，其高度和为中心高度的两倍。

3. 底面积乘平均高度，并检查高度在整个底面非负。

> Example. Above the disk $x^2 + y^2 \le 4$, a solid lies between $z = 0$ and $z = 3 + x$. Find its volume.

> 例题：立体位于圆盘 $x^2 + y^2 \le 4$ 上方、$z = 0$ 与 $z = 3 + x$ 之间，求体积。

Worked solution

1. The height is nonnegative because $-2 \le x \le 2$, so $1 \le 3 + x \le 5$.

2. Opposite disk points have heights $3 + x$ and $3 - x$, averaging to 3.

3. The disk area is 4π, so volume is 12π.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 6 / 方法 6 [GE10-M06] (continued / 续)

例题解析

1. 由于 $-2 \le x \le 2$，高度介于 1 与 5，始终非负。

2. 相反圆盘点的高度为 $3+x, 3-x$，平均值为 3。

3. 底面积为 4π，所以体积为 12π。

If height changes sign, the desired solid needs a separate region split.

若高度变号，所求立体必须另行分区。

Practice / 配套练习: AIME-2003-II-Q05 (WB p. 439)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 7 / 方法 7 [GE10-M07]

Visibility and paint depend on a cube cell’s boundary position

可见性与油漆取决于单位格的边界位置

Procedure

1. Encode whether a cell touches each designated outer face.

2. Count hidden cells with shifted dimensions, such as $(p-1)(q-1)(r-1)$ for three adjacent visible faces.

3. For exposed or unpainted face pairs, classify corner, edge, face-interior and solid-interior positions before counting orientations.

通用步骤

1. 记录单位格是否接触各指定外表面。

2. 用平移后的尺寸计隐藏格，例如三个相邻可见面对应 $(p-1)(q-1)(r-1)$。

3. 计外露或未染面组合时，先按角、棱、面内及体内位置分类，再计朝向。

> Example. A $3\times4\times5$ block of unit cubes is viewed with three adjacent faces visible. How many cubes touch none of those faces?

> 例题：从三个相邻面观察 $3\times4\times5$ 单位立方体块，有多少小立方体完全不接触这些面?

Worked solution

1. In each coordinate direction exclude the first layer touching the corresponding visible face.

2. The remaining index choices are $2,3,4$ independently.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 7 / 方法 7 [GE10-M07] (continued / 续)

3. Thus the hidden count is $2 \cdot 3 \cdot 4 = 24$. This is different from removing all six outer layers.

例题解析

1. 每个坐标方向排除与相应可见面相接的首层。

2. 三个方向独立剩余 $2, 3, 4$ 个指标。

3. 所以隐藏格数为 24；这不同于移除全部六个外表面层。

Visible from three faces does not mean exposed on any of six faces.

从三个面可见，不等于是否接触六个表面中的任意一个。

Two adjacent marked cube faces meet along one of twelve equally likely edges. To keep both inward, count unordered inward-face pairs minus opposite-face pairs. For a corner, edge position and face center these counts are $\binom{3}{2} = 3$, $\binom{4}{2} - 1 = 5$ and $\binom{5}{2} - 2 = 8$, respectively, each out of twelve.

立方体两个相邻标记面交于十二条等可能棱之一。要使两面均朝内，可计向内面的无序二元组，再扣除对面组合。大立方体角格、棱格、面心格的合法数分别为 $\binom{3}{2} = 3$, $\binom{4}{2} - 1 = 5$, $\binom{5}{2} - 2 = 8$，总朝向类别均为十二。

Practice / 配套练习: AIME-2004-II-Q03 (WB p. 531), AIME-2005-I-Q09 (WB p. 737)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08]

Develop curved surfaces before measuring paths or rolling

测量路径或滚动前先展开曲面

Procedure

1. A cylinder develops to a rectangle; a cone of slant height ℓ and base radius r develops to angle $2\pi r/\ell$.

2. Taut geodesics become straight lines in a valid development; compare angular lifts or winding classes.

3. For rolling, equate no-slip arc lengths and then recover the desired height or length.

4. At a tangent departure from a cylinder, preserve the developed line's pitch in the free segment.

通用步骤

1. 圆柱展开为矩形；母线长 ℓ、底半径 r 的圆锥展开角为 $2\pi r/\ell$。

2. 绷紧最短路径在合法展开中成为直线，并须比较绕行或角度提升类别。

3. 对无滑动滚动令对应弧长相等，再恢复目标高度或长度。

4. 从圆柱相切离开的自由线段，须保持展开直线的坡度。

> Example. A cylinder has circumference 8 and height 3. Find the shortest path on its lateral surface from a point on its bottom rim to the diametrically opposite point on its top rim. How many shortest paths are there?

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08] (continued / 续)

例题：圆柱周长为 8、高为 3。从下底圆周一点沿侧面到上底圆周上与其对径的位置，求最短距离及最短路径条数。

Worked solution

1. Unroll the lateral surface into its periodic strip. Put the start at $(0,0)$; the endpoint lifts are $(4+8k,3)$ for integers k.

2. A straight lifted path has squared length $(4+8k)^2+9$. The least absolute horizontal displacement is four, attained exactly at $k=0,-1$.

3. The minimum distance is therefore 5. Both straight segments stay between the two rim lines, giving two shortest paths, one around each side of the cylinder.

例题解析

1. 把侧面展开为周期条带。起点取 $(0,0)$，终点各提升为 $(4+8k,3)$，其中 k 为整数。

2. 直线提升的长度平方为 $(4+8k)^2+9$。最小水平位移绝对值为四，且仅在 $k=0,-1$ 达到。

3. 所以最短距离为 5；两条直线均在上下底边之间，投回圆柱后得到分别绕两侧的两条最短路径。

A planar angle between generators is not generally their developed angle.

母线的空间角通常不等于展开后的角。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08] (continued / 续)

For a rolling cone completing one circuit about its fixed vertex, the contact travel is $2\pi\ell$, equal to the number of axial spins times $2\pi r$. For a cylinder rope, the free tangent's horizontal length is $\sqrt{d^2-R^2}$; use the common length-to-horizontal-projection ratio.

圆锥绕固定顶点滚完一圈，接触路程为 $2\pi\ell$，等于轴向自转圈数乘 $2\pi r$。圆柱绳索自由切线的水平长度为 $\sqrt{d^2-R^2}$，再用共同的实际长度与水平投影比。

Practice / 配套练习: AIME-2004-I-Q14 (WB p. 454), AIME-2004-II-Q11 (WB p. 455), AIME-2008-I-Q05 (WB p. 440)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09]

Find the perpendicular height before using a solid-volume formula

套用立体体积公式前先求垂直高度

Procedure

1. Distinguish edge length, face slant height and perpendicular solid height.

2. Use a right section, base circumradius or explicit center coordinates to recover that height.

3. Apply $V = Kh/3$ for a pyramid, or a dimension product for a box; only then simplify or optimize.

通用步骤

1. 区分棱长、面上斜高及立体的垂直高度。

2. 用直角截面、底面外接圆半径或明确圆心坐标求垂直高。

3. 棱锥用 $V = Kh/3$，长方体用三维乘积；最后再化简或优化。

> Example. A tetrahedron has a right-triangle base with legs $6, 8$. Each edge from its apex to a base vertex has length 13. Find its volume.

> 例题：四面体底面为两直角边长 $6, 8$ 的直角三角形，顶点到三个底面顶点的距离均为 13。求体积。

Worked solution

1. The base hypotenuse is ten, so its circumcenter is the hypotenuse midpoint and its circumradius is five. Its area is $6 \cdot 8/2 = 24$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09] (continued / 续)

2. Equal apex distances put its perpendicular projection at that circumcenter. The vertical height is $\sqrt{13^2-5^2}=12$.

3. Thus the volume is $24\cdot 12/3=96$. Placing the apex twelve units above the circumcenter realizes all three required lateral lengths.

例题解析

1. 底面斜边为十，外心为斜边中点，外接圆半径为五，面积为 $6\cdot 8/2=24$。

2. 顶点到三底面点等距，故其垂足在外心；垂直高为 $\sqrt{13^2-5^2}=12$。

3. 体积为 $24\cdot 12/3=96$。将顶点放在外心正上方十二处，确能实现三条长为十三的侧棱。

Equal lateral edges put the apex over the base circumcenter, not necessarily its centroid.

侧棱相等使顶点在底面外心上方，不一定在重心上方。

For the regular octahedron with axial vertices $(\pm1,0,0),(0,\pm1,0),(0,0,\pm1)$, two square pyramids give volume $4/3$. Its face centers are $(\pm1/3,\pm1/3,\pm1/3)$, a cube of side $2/3$.

If box cuts preserve a fixed sum of the remaining dimensions, AM–GM bounds the volume; confirm the equal-dimension candidate uses an allowed integral number of cuts.

轴上顶点为 $(\pm1,0,0),(0,\pm1,0),(0,0,\pm1)$ 的正八面体，可由两个正四棱锥求得体积 $4/3$。各面重心为 $(\pm1/3,\pm1/3,\pm1/3)$，形成边长 $2/3$ 的立方体。

若切割长方体后剩余三维和固定，可由均值不等式限制体积；还须确认等边长候选对应合法的整数切片数。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09] (continued / 续)

Practice / 配套练习: AIME-2005-II-Q10 (WB p. 446), AIME-2008-II-Q03 (WB p. 181), AIME-2012-II-Q05 (WB p. 441), AIME-2017-I-Q04 (WB p. 443), AIME-2022-II-Q03 (WB p. 423)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 10 / 方法 10 [GE10-M10]

Planar section area is projected area divided by a cosine

平面截面积等于投影面积除以余弦

Procedure

1. Find the actual section polygon by intersecting the cutting plane with solid edges or curved boundaries.

2. Project to a convenient coordinate plane, retaining exactly the allowed projected region.

3. If the section plane is $z = px + qy + c$, multiply projected area by $\sqrt{1 + p^2 + q^2}$.

通用步骤

1. 将截平面与立体各棱或曲边相交，确定实际截面多边形。

2. 向方便的坐标面投影，精确保留合法投影区域。

3. 若截平面为 $z = px + qy + c$，则投影面积乘 $\sqrt{1 + p^2 + q^2}$。

> Example. A plane $z = x + y$ cuts the vertical prism over the unit square, with enough height to contain the entire patch. Find the patch area.

> 例题：平面 $z = x + y$ 截过以单位正方形为底的直棱柱，棱柱高度足以包含整片截面，求截片面积。

Worked solution

1. The four section vertices are $(0, 0, 0), (1, 0, 1), (1, 1, 2), (0, 1, 1)$.

2. Their horizontal projection is the unit square of area one. Here $p = q = 1$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 10 / 方法 10 [GE10-M10] (continued / 续)

3. The section area is $1\sqrt{1+1+1}=\sqrt{3}$. The same value follows from the cross product of adjacent section edges.

例题解析

1. 四个截面顶点为 $(0,0,0),(1,0,1),(1,1,2),(0,1,1)$。

2. 水平投影为面积一的单位正方形，且 $p=q=1$。

3. 实际面积为 $\sqrt{1+1+1}=\sqrt{3}$，亦可由相邻截面边叉积核实。

An infinite-plane ellipse is not always the cylinder’s bounded section: top and bottom caps may truncate it.

无限平面与圆柱面的椭圆，不一定就是有限圆柱截面；上下底面可能将它截断。

A tilted plane through a cylinder center can project to a disk strip. Compute the strip by subtracting circular caps, then apply the tilt factor; do not replace the strip by a full ellipse.

过圆柱中心的斜平面可能投影为圆盘条带，应先扣除圆弓形计算条带，再乘倾斜系数，不可直接当成完整椭圆。

Practice / 配套练习: AIME-2007-I-Q13 (WB p. 456), AIME-2015-I-Q15 (WB p. 459)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 11 / 方法 11 [GE10-M11]

Fold rigid panels by matching their projected seam points

通过拼缝投影相合求刚性侧板折叠

Procedure

1. Choose coordinates in the original sheet and locate the hinge lines and cut vertices.

2. After folding, distances to a hinge line remain fixed; symmetry determines which seam points must coincide.

3. Split the invariant panel distance into horizontal and vertical components and solve for height.

通用步骤

1. 在原纸面取坐标，定位折线与切口顶点。

2. 折叠后到铰链直线的距离保持不变，对称性确定必须重合的拼缝点。

3. 把不变的侧板距离分成水平与竖直分量，求高度。

> Example. Two perpendicular hinge lines are $x=3, y=3$. Points $(2,0)$ and $(0,2)$ on their respective symmetric panels are folded up and joined above $(2,2)$. Find the common height.

> 例题：两条垂直折线为 $x=3, y=3$。对称侧板上的点 $(2,0),(0,2)$ 折起后在 $(2,2)$ 正上方拼合，求高度。

Worked solution

1. The point $(2,0)$ is distance three from its hinge $y=3$. Its perpendicular hinge foot is $(2,3)$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 11 / 方法 11 [GE10-M11] (continued / 续)

2. After joining, its horizontal separation from that foot is one. Rigid folding keeps the total separation three.

3. Thus $h^2 + 1^2 = 3^2$, giving $h = 2\sqrt{2}$. The other panel gives the same height by reflection.

例题解析

1. 点 $(2, 0)$ 到折线 $y = 3$ 的距离为三，垂足为 $(2, 3)$。

2. 拼合后其到垂足的水平距离为一，刚性折叠使总距离仍为三。

3. 故 $h^2 + 1^2 = 3^2$，得 $h = 2\sqrt{2}$；另一侧板由对称给出同高。

Folding a tray does not imply its walls become vertical.

托盘折起后，其侧壁不一定竖直。

Practice / 配套练习: AIME-2008-I-Q15 (WB p. 457)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 12 / 方法 12 [GE10-M12]

Revolve a triangle as two cones sharing a base

将三角形旋转分成共底双圆锥

Procedure

1. Resolve all planar inequalities to locate the triangular region and the rotation axis.

2. Project the off-axis vertex perpendicularly and check whether its foot lies on the base segment.

3. If the foot is internal, the two cones share radius equal to that perpendicular distance; their heights sum to the base length.

通用步骤

1. 先解平面不等式，确定三角形区域及旋转轴。

2. 从轴外顶点作垂足，并检查垂足是否在底边线段内。

3. 若垂足在内部，两圆锥共底半径为该垂距，其高之和为底边长。

> Example. Revolve triangle $A=(-2,0), B=(4,0), C=(1,3)$ about the x axis. Find its volume.

> 例题：将三角形 $A=(-2,0), B=(4,0), C=(1,3)$ 绕 x 轴旋转，求体积。

Worked solution

1. The perpendicular foot of C is $(1,0)$, inside AB. Both cone radii are three.

2. The left and right heights are three and three, summing to six.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 12 / 方法 12 [GE10-M12] (continued / 续)

3. Total volume is $\pi \cdot 3^2(3+3)/3 = 18\pi$. Their interiors meet only at the common base.

例题解析

1. C 的垂足为 $(1,0)$，位于 AB 内，两圆锥半径均为三。

2. 左右高分别为三，总高为六。

3. 总体积为 $\pi \cdot 3^2(3+3)/3 = 18\pi$；两块仅在共底边界接触。

If the foot is outside, adding two cones double-counts; a difference model is required.

若垂足在外，相加会重复计算，应改用圆锥差。

Practice / 配套练习: AIME-2010-I-Q11 (WB p. 458)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 13 / 方法 13 [GE10-M13]

Clip a first-octant tetrahedron by box constraints

用长方体限制截割第一卦限四面体

Procedure

1. Write the cutting plane in edge coordinates and choose the desired half-space.

2. Use the first-octant simplex volume $abc/6$ from its three intercepts.

3. Subtract portions beyond the box faces, using translated intercepts and inclusion–exclusion when needed.

4. For a tilted immersed cube, derive the height functional from orthonormal edge projections before clipping.

通用步骤

1. 在棱坐标中写出截平面，并选择目标半空间。

2. 由三轴截距 a, b, c 使用第一卦限四面体体积 $abc/6$。

3. 扣除超出长方体各面的部分，使用平移截距，必要时作容斥。

4. 倾斜浸水立方体应先由正交棱投影求出高度函数，再进行截割。

> Example. Find the volume in $[0,2]^3$ satisfying $x+y+z<3$.

> 例题：求 $[0,2]^3$ 内满足 $x+y+z<3$ 的部分体积。

Worked solution

1. The full first-octant tetrahedron has intercepts three, so volume $27/6$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 13 / 方法 13 [GE10-M13] (continued / 续)

2. Beyond each of $x = 2, y = 2, z = 2$, translate that coordinate by two; the excluded simplex has intercepts one and volume $1/6$.

3. No point can exceed two coordinates of two while their sum is below three, so there are no overlapping exclusions. Volume is $27/6 - 3/6 = 4$.

例题解析

1. 完整第一卦限四面体三截距均为三，体积为 $27/6$。

2. 超出 $x = 2, y = 2, z = 2$ 的每块，分别平移相应坐标二，得到截距均一、体积 $1/6$ 的四面体。

3. 坐标和小于三时，不可能有两个坐标同时大于二，因此排除块不重叠，体积为 4。

Check every coordinate ceiling; not all three truncations necessarily occur.

各坐标上界须分别检查，不一定三个方向都发生截割。

For a cube with a body diagonal vertical, a near-corner horizontal section has equal edge intercepts and is equilateral. Verify those intercepts are below the edge length and that the submerged depth is less than the container depth.

立方体体对角线竖直时，靠近角的水平截面具有相等棱截距，为等边三角形。须检查截距小于棱长，并检查浸没深度小于容器深度。

Practice / 配套练习: AIME-2012-I-Q08 (WB p. 447), AIME-2015-II-Q09 (WB p. 448), AIME-2023-II-Q14 (WB p. 461)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 14 / 方法 14 [GE10-M14]

Dihedral angles live in a perpendicular section

二面角应在垂直截面中测量

Procedure

1. Locate the line where the two planes meet.

2. Choose one line in each plane perpendicular to that common line at the same point.

3. The angle between these section lines is the dihedral angle; use planar trigonometry to recover spatial height.

通用步骤

1. 找出两平面的公共交线。

2. 在交线同一点，分别于两平面内作垂直交线的直线。

3. 这两截线夹角才是二面角，再用平面三角法求空间高度。

> Example. A triangular base has altitude 4 from vertex A to side UV. A point W is directly h above A. If planes AUV, WUV form 60°, find h.

> 例题：三角形底面中 A 到边 UV 的高为四，点 W 位于 A 正上方高 h 处。若平面 AUV, WUV 的二面角为 60°，求 h。

Worked solution

1. Let M be the perpendicular foot from A to UV. Since AW is normal to the base, WM is also perpendicular to UV.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 14 / 方法 14 [GE10-M14] (continued / 续)

2. The required section angle is $\angle AMW = 60^\circ$. Triangle AMW is right at A.

3. Thus $h = AM\tan 60^\circ = 4\sqrt{3}$.

例题解析

1. 令 M 为 A 到 UV 的垂足；因 AW 垂直底面，WM 也垂直 UV。

2. 所需截面角为 $\angle AMW = 60^\circ$，而 AMW 在 A 为直角。

3. 所以 $h = 4\tan 60^\circ = 4\sqrt{3}$。

Two arbitrary lines in the planes do not measure their dihedral.

分别取两平面内任意两条线，不能测得二面角。

Practice / 配套练习: AIME-2016-I-Q04 (WB p. 442), AIME-2016-II-Q14 (WB p. 433)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 15 / 方法 15 [GE10-M15]

Face diagonals determine a Gram volume

面对角线确定格拉姆体积

Procedure

1. Choose generating edge vectors u, v, w and use $|u \pm v|^2$ to recover edge norms and dot products.

2. Form their Gram matrix, whose determinant equals squared parallelepiped volume.

3. Keep all feasible dot-product sign choices; the sign of the triple product of pairwise dots can change volume.

通用步骤

1. 取生成棱向量 u, v, w，用 $|u \pm v|^2$ 求棱范数与点积。

2. 组成格拉姆矩阵，其行列式等于平行六面体体积平方。

3. 保留全部可行点积符号组合；三个两两点积之积的符号可能改变体积。

> Example. Unit vectors u, v, w satisfy $u \cdot v = u \cdot w = v \cdot w = 1/2$. Find the parallelepiped volume.

> 例题：单位向量 u, v, w 的两两点积均为 $1/2$，求所生成平行六面体体积。

Worked solution

1. The Gram matrix has diagonal entries one and off-diagonal entries $1/2$.

2. Its determinant is $1 - 3(1/2)^2 + 2(1/2)^3 = 1/2$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 15 / 方法 15 [GE10-M15] (continued / 续)

3. The volume is $1/\sqrt{2}$. Positive determinant together with positive leading principal minors confirms a nondegenerate realization.

例题解析

1. 格拉姆矩阵对角元均一，非对角元均 $1/2$。

2. 行列式为 $1-3(1/2)^2+2(1/2)^3=1/2$。

3. 体积为 $1/\sqrt{2}$；各顺序主子式均正，确认存在非退化空间实现。

An algebraic dot-product pattern needs a positive semidefinite Gram matrix to be geometrically feasible.

代数点积模式需有半正定格拉姆矩阵，才能在几何上实现。

Practice / 配套练习: AIME-2023-I-Q13 (WB p. 460)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16]

Opposite equal edges embed a tetrahedron in a box

对棱相等可把四面体嵌入长方体

Procedure

1. Use alternate vertices of a box so opposite tetrahedron edges are diagonals of corresponding box faces.

2. Solve the three pair-sum equations for squared box dimensions and require positivity.

3. Compute volume as one third of box volume; compute face area and use $3V = rS$ for an insphere.

通用步骤

1. 取长方体相间顶点，使四面体对棱成为相应长方体面的对角线。

2. 解三组平方和方程求长方体边长平方，并要求均为正。

3. 四面体体积为长方体的三分之一；求各面面积后，用 $3V = rS$ 求内切球。

> Example. Could a tetrahedron have its three opposite-edge pairs of lengths $\sqrt{2}, \sqrt{3}, \sqrt{6}$? Each proposed face is a valid triangle; determine whether the spatial solid exists.

> 例题：四面体三组对棱长度能否分别为 $\sqrt{2}, \sqrt{3}, \sqrt{6}$？虽然每个拟定面都满足三角不等式，请判断空间立体是否存在。

Worked solution

1. The strongest face inequality is $\sqrt{2} + \sqrt{3} > \sqrt{6}$, so planar face validity alone gives no obstruction.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16] (continued / 续)

2. For an opposite-edge-equal tetrahedron, box dimensions would satisfy $a^2+b^2=2, a^2+c^2=3, b^2+c^2=6$.

3. The first two equations minus the third give $2a^2=-1$, impossible for a real box dimension. Thus the tetrahedron does not exist, despite the valid individual faces.

例题解析

1. 最强的面三角不等式为 $\sqrt{2}+\sqrt{3}>\sqrt{6}$，所以单独检查各面无法排除。

2. 若对棱相等四面体存在，长方体边长应满足 $a^2+b^2=2, a^2+c^2=3, b^2+c^2=6$。

3. 前两式相加再减第三式得 $2a^2=-1$，不可能对应实数边长。因此各面虽合法，空间四面体仍不存在。

If exceptional edge lengths occur in pairs of faces, first establish which edges are opposite.

若特殊棱长通过各面条件给出，应先确定哪些棱互为对棱。

For face squared sides A, B, C, $16K^2=2(AB+BC+CA)-(A^2+B^2+C^2)$. When all four faces are congruent, total surface area is $4K$.

The box representation is a theorem, not an extra assumption. Put the tetrahedron centroid at zero and let u, v, w be the midpoints of the three edges from one vertex. The four vertex vectors are $u+v+w, u-v-w, -u+v-w, -u-v+w$. Equality of each opposite-edge pair gives $|v+w|=|v-w|$ and its cyclic counterparts, so u, v, w are pairwise perpendicular. Their doubled lengths are the box dimensions; nondegeneracy requires all three to be positive.

若一面的三边平方为 A, B, C，则 $16K^2=2(AB+BC+CA)-(A^2+B^2+C^2)$；四面全等时总表面积为 $4K$。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16] (continued / 续)

长方体表示是定理而非额外假设。把四面体重心放在原点，令 u, v, w 为同一顶点引出的三条棱的中点向量，则四顶点向量为 $u+v+w, u-v-w, -u+v-w, -u-v+w$。每组对棱相等给出 $|v+w| = |v-w|$ 及循环同式，所以 u, v, w 两两垂直。其长度的两倍即长方体三边，非退化要求三者均正。

Practice / 配套练习: AIME-2024-I-Q14 (WB p. 462)